

Comp-2540: Dynamic Programming part 2: Edit Distance

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"Those who cannot remember the past are condemned to repeat it."

Outline

- 1 Algorithm design strategies

Algorithm design strategies

- Generate and test
- Greedy method
- Divide and conquer
- Dynamic programming

Dynamic Programming

- Used on two types of problems:
 - Optimization problems: select a feasible solution, so that the value of the required function is minimized or maximized.
 - Combinatorial problems: figure out the number of ways to do something
- Divide-and-conquer: It divides a problem into sub-problems. (so it is an extension of Divide and Conquer)
- Overlapping sub-solutions: The sub-solutions overlap. These sub-solutions are computed over and over again when computing the global optimal solution in a brute-force algorithm.
- Memorization: Use memorization to save repeated solution for the same subproblem.
- Optimal substructure: The optimal solution to the problem contains within its subsolutions.

- The edit distance between string X and string Y is the minimum number of basic operations that convert X to Y .
- Applications
 - Spelling correction
 - DNA sequence similarities
 - Foundation for other algorithms (e.g., Longest sub sequence problem).
- Levenshtein distance: The admissible basic operations are
 - insert, cost =1
 - delete, cost =1
 - replace, cost=1
 - copy, cost=0
- Levenshtein distance

X	Y	operation(s)	cost
dog	do	delete	1
cat	cart	insert	1
cat	cut	replace	1
cat	act	delete, insert	2

Optimal solution for edit distance

Given *cats* and *fast*, what are the solutions for their edit distance?

- Solution 1: delete four letters c, a, t, s, then add four letters f, a, s, t
 - cost= 8
- Solution 2: replace c by f, copy a, replace t by s and s by t
 - cost= 3
- ...
- There are many solutions
- What we are interested in is to find the optimal solution
- This kind of problem is called an optimization problem

How to find the optimal solution: Divide and Conquer

- Break the original problem into a smaller problem.
- By removing the last letter of *cats* and *fast*.

*cat***s** $\longrightarrow$ *fa***t**

The last letter can be transformed in 3 cases:

- 1 replace. The subproblem is (cat, fas)
- 2 insert **t**: the subproblem is (cats, fas)
- 3 delete **s**: the subproblem is (cat, fast)

problem definition

- For two strings
 - X of length n
 - Y of length m
- $D(i,j)$
 - the edit distance between $X[1..i]$ and $Y[1..j]$
 - score of the best alignment from $X[1..i]$ to $Y[1..j]$
 - i.e., the first i characters of X and the first j characters of Y
 - The edit distance between X and Y is thus $D(n,m)$
- Properties for $D(i,j)$
 - $D(i,0)=i$; delete i letters
 - $D(0,j)=j$; insert j letters

recurrence relation

$$D(i, j) = \begin{cases} D(i-1, j-1) + d(X_i, Y_j); & \text{replace or copy} \\ D(i-1, j) + 1; & \text{insert} \\ D(i, j-1) + 1; & \text{delete} \end{cases} \quad (1)$$

$$d(x, y) = \begin{cases} 0; & \text{if } x=y \\ 1; & \text{otherwise} \end{cases} \quad (2)$$

Edit distance using dynamic programming

- Dynamic programming: A tabular computation of $D(n,m)$
- Solving problems by combining solutions to subproblems.
- Bottom-up
 - We compute $D(i,j)$ for small i,j
 - And compute larger $D(i,j)$ based on previously computed smaller values
 - i.e., compute $D(i,j)$ for all i ($0 < i < n$) and j ($0 < j < m$)

Levenshtein distance: Computation

		f	a	s	t
	0	1	2	3	4
c	1	1	2	3	4
a	2	2	1	2	3
t	3	3	2	2	2
s	4	4	3	2	3

Levenshtein distance: Algorithm

LEVENSHTEINDISTANCE(s_1, s_2)

```

1  for  $i \leftarrow 0$  to  $|s_1|$ 
2  do  $m[i, 0] = i$ 
3  for  $j \leftarrow 0$  to  $|s_2|$ 
4  do  $m[0, j] = j$ 
5  for  $i \leftarrow 1$  to  $|s_1|$ 
6  do for  $j \leftarrow 1$  to  $|s_2|$ 
7      do if  $s_1[i] = s_2[j]$ 
8          then  $m[i, j] = \min\{m[i-1, j]+1, m[i, j-1]+1, m[i-1, j-1]\}$ 
9          else  $m[i, j] = \min\{m[i-1, j]+1, m[i, j-1]+1, m[i-1, j-1]+1\}$ 
10 return  $m[|s_1|, |s_2|]$ 

```

Operations: insert (cost 1), delete (cost 1), replace (cost 1), copy (cost 0)

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```

Operations: insert (cost 1), delete (cost 1), replace (cost 1), copy (cost 0)

Levenshtein distance: Example

		f		a		s		t	
		<u>0</u>	<u>1</u> <u>1</u>	<u>2</u> <u>2</u>	<u>3</u> <u>3</u>	<u>4</u> <u>4</u>			
c		<u>1</u>	<u>1</u> <u>2</u>	<u>2</u> <u>3</u>	<u>3</u> <u>4</u>	<u>4</u> <u>5</u>			
		<u>1</u>	<u>2</u> <u>1</u>	<u>2</u> <u>2</u>	<u>3</u> <u>3</u>	<u>4</u> <u>4</u>			
a		<u>2</u>	<u>2</u> <u>2</u>	<u>1</u> <u>3</u>	<u>3</u> <u>4</u>	<u>4</u> <u>5</u>			
		<u>2</u>	<u>3</u> <u>2</u>	<u>3</u> <u>1</u>	<u>2</u> <u>2</u>	<u>3</u> <u>3</u>			
t		<u>3</u>	<u>3</u> <u>3</u>	<u>3</u> <u>2</u>	<u>2</u> <u>3</u>	<u>2</u> <u>4</u>			
		<u>3</u>	<u>4</u> <u>3</u>	<u>4</u> <u>2</u>	<u>3</u> <u>2</u>	<u>3</u> <u>2</u>			
s		<u>4</u>	<u>4</u> <u>4</u>	<u>4</u> <u>3</u>	<u>2</u> <u>3</u>	<u>3</u> <u>3</u>			
		<u>4</u>	<u>5</u> <u>4</u>	<u>5</u> <u>3</u>	<u>4</u> <u>2</u>	<u>3</u> <u>3</u>			

- Bold font: minimum

Each cell of Levenshtein matrix

cost of getting here from my upper left neighbor (copy or replace)	cost of getting here from my upper neighbor (delete)
cost of getting here from my left neighbor (insert)	the minimum of the three possible “movements”; the cheapest way of getting here

Tracing the algorithm

- 1 Compute Levenshtein distance matrix for OSLO – SNOW
- 2 What are the Levenshtein editing operations that transform *cat* into *catcat*?

Transform *OSLO* to *SNOW*

- Column 0: cost of deleting *oslo*
- Row 0: cost of inserting *snow*

		s	n	o	w
	0	1 1	2 2	3 3	4 4
o	1 1				
s	2 2				
l	3 3				
o	4 4				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{1}{\quad} 1$	$\frac{2}{\quad} 2$	$\frac{3}{\quad} 3$	$\frac{4}{\quad} 4$
o	$\frac{1}{\quad} 1$	$\frac{1}{2} \frac{2}{?}$			
s	$\frac{2}{\quad} 2$				
l	$\frac{3}{\quad} 3$				
o	$\frac{4}{\quad} 4$				

$o \rightarrow s$

Can be achieved in three different ways:

- ① if o is already removed: insert s . The subproblem is $EMPTY \rightarrow s$
- ② if s is already inserted, remove o . The subproblem is $EMPTY \rightarrow s$
- ③ or we can simply replace o with s . The subproblem is $EMPTY \rightarrow EMPTY$

			s	n	o	w
		0	1 1	2 2	3 3	4 4
o		1 1	1 2 2 1			
s		2 2				
l		3 3				
o		4 4				

$o \rightarrow sn$ can be achieved in three different ways:

- 1 subproblem is (o, s) : insert n .

$oslo \rightarrow \text{replace } o \text{ with } s \text{ } \textcolor{red}{s}slo \rightarrow \text{insert } n \text{ } \textcolor{red}{snslo}$

- 2 subproblem is $(EMPTY, sn)$: remove o .

$oslo \rightarrow \text{insert } s \text{ } n \text{ } \textcolor{red}{snoslo} \rightarrow \text{delete } o \text{ } \textcolor{red}{snslo}$

- 3 subproblem is $(EMPTY, s)$: replace o by n

$oslo \rightarrow \text{insert } s \text{ } \textcolor{red}{soslo} \rightarrow \text{replace } o \text{ by } n \text{ } \textcolor{red}{snslo}$

			s	n	o	w
		0	1 1	2 2	3 3	4 4
o		1 1	1 2 2 1	2 3 2 ?		
s		2 2				
l		3 3				
o		4 4				

Take the minimum among the three solutions

		s	n	o	w
	$\frac{\quad}{0}$	$\frac{1}{1}$	$\frac{2}{2}$	$\frac{3}{3}$	$\frac{4}{4}$
o	$\frac{1}{1}$	$\frac{1}{2}$	$\frac{2}{2}$		
s	$\frac{2}{2}$				
l	$\frac{3}{3}$				
o	$\frac{4}{4}$				

$o \rightarrow sno$

- 1 subproblem is (o, sn). Insert o.

$oslo \xrightarrow{\text{replace } o \text{ with } s} sslo \xrightarrow{\text{insert } n} snslo \rightarrow snoslo$

- 2 subproblem is (EMPTY, sno): remove o.

$oslo \xrightarrow{\text{insert } s \text{ n } o} snooslo \xrightarrow{\text{delete } o} snoslo$

- 3 subproblem is (EMPTY, sn): copy o

$oslo \xrightarrow{\text{insert } s \text{ n}} snoslo \xrightarrow{\text{copy } o} snoslo$

		s	n	o	w
	0	1 1	2 2	3 3	4 4
o	1 1	1 2 2 1	2 3 2 2	2 4 3 ?	
s	2 2				
l	3 3				
o	4 4				

		s	n	o	w
	$\frac{\quad}{0}$	$\frac{1}{1}$	$\frac{2}{2}$	$\frac{3}{3}$	$\frac{4}{4}$
o	$\frac{1}{1}$	$\frac{1}{2}$ $\frac{2}{1}$	$\frac{2}{2}$ $\frac{3}{2}$	$\frac{2}{3}$ $\frac{4}{2}$	
s	$\frac{2}{2}$				
l	$\frac{3}{3}$				
o	$\frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{?}$
s	$\frac{\quad}{\quad} \frac{2}{2}$				
l	$\frac{\quad}{\quad} \frac{3}{3}$				
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$				
l	$\frac{\quad}{\quad} \frac{3}{3}$				
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{?}$			
l	$\frac{\quad}{\quad} \frac{3}{3}$				
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$			
l	$\frac{\quad}{\quad} \frac{3}{3}$				
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{0}$	$\frac{1}{1}$	$\frac{2}{2}$	$\frac{3}{3}$	$\frac{4}{4}$
o	$\frac{1}{1}$	$\frac{1}{2} \frac{2}{1}$	$\frac{2}{2} \frac{3}{2}$	$\frac{2}{3} \frac{4}{2}$	$\frac{4}{3} \frac{5}{3}$
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l	$\frac{3}{3}$				
o	$\frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$		
l	$\frac{\quad}{\quad} \frac{3}{3}$				
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{0}$	$\frac{1}{1}$	$\frac{2}{2}$	$\frac{3}{3}$	$\frac{4}{4}$
o	$\frac{1}{1}$	$\frac{1}{2} \frac{2}{1}$	$\frac{2}{2} \frac{3}{2}$	$\frac{2}{3} \frac{4}{2}$	$\frac{4}{3} \frac{5}{3}$
s	$\frac{2}{2}$	$\frac{1}{3} \frac{2}{1}$	$\frac{2}{2} \frac{3}{2}$	$\frac{3}{3} \frac{3}{?}$	
l	$\frac{3}{3}$				
o	$\frac{4}{4}$				

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	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{3}{3}$	
l	$\frac{\quad}{\quad} \frac{3}{3}$				
o	$\frac{\quad}{\quad} \frac{4}{4}$				

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s	$\frac{2}{2}$	$\frac{1}{3} \frac{2}{1}$	$\frac{2}{2} \frac{3}{2}$	$\frac{3}{3} \frac{3}{3}$	$\frac{3}{4} \frac{4}{?}$
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s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{4}{3}$
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s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{4}{3}$
l	$\frac{\quad}{\quad} \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{2}{?}$			
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	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{4}{3}$
l	$\frac{\quad}{\quad} \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{2}{2}$			
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{0}$	$\frac{\quad}{1} \quad \frac{\quad}{1}$	$\frac{\quad}{2} \quad \frac{\quad}{2}$	$\frac{\quad}{3} \quad \frac{\quad}{3}$	$\frac{\quad}{4} \quad \frac{\quad}{4}$
o	$\frac{\quad}{1} \quad \frac{\quad}{1}$	$\frac{\quad}{1} \quad \frac{\quad}{2} \quad \frac{\quad}{2} \quad \frac{\quad}{1}$	$\frac{\quad}{2} \quad \frac{\quad}{3} \quad \frac{\quad}{2} \quad \frac{\quad}{2}$	$\frac{\quad}{2} \quad \frac{\quad}{4} \quad \frac{\quad}{3} \quad \frac{\quad}{2}$	$\frac{\quad}{4} \quad \frac{\quad}{5} \quad \frac{\quad}{3} \quad \frac{\quad}{3}$
s	$\frac{\quad}{2} \quad \frac{\quad}{2}$	$\frac{\quad}{1} \quad \frac{\quad}{2} \quad \frac{\quad}{3} \quad \frac{\quad}{1}$	$\frac{\quad}{2} \quad \frac{\quad}{3} \quad \frac{\quad}{2} \quad \frac{\quad}{2}$	$\frac{\quad}{3} \quad \frac{\quad}{3} \quad \frac{\quad}{3} \quad \frac{\quad}{3}$	$\frac{\quad}{3} \quad \frac{\quad}{4} \quad \frac{\quad}{4} \quad \frac{\quad}{3}$
l	$\frac{\quad}{3} \quad \frac{\quad}{3}$	$\frac{\quad}{3} \quad \frac{\quad}{2} \quad \frac{\quad}{4} \quad \frac{\quad}{2}$	$\frac{\quad}{2} \quad \frac{\quad}{3} \quad \frac{\quad}{3} \quad \frac{\quad}{?}$		
o	$\frac{\quad}{4} \quad \frac{\quad}{4}$				

		s	n	o	w
	$\frac{\quad}{0}$	$\frac{\quad}{1} \quad \frac{\quad}{1}$	$\frac{\quad}{2} \quad \frac{\quad}{2}$	$\frac{\quad}{3} \quad \frac{\quad}{3}$	$\frac{\quad}{4} \quad \frac{\quad}{4}$
o	$\frac{\quad}{1} \quad \frac{\quad}{1}$	$\frac{\quad}{2} \quad \frac{\quad}{1}$	$\frac{\quad}{2} \quad \frac{\quad}{2}$	$\frac{\quad}{3} \quad \frac{\quad}{2}$	$\frac{\quad}{3} \quad \frac{\quad}{3}$
s	$\frac{\quad}{2} \quad \frac{\quad}{2}$	$\frac{\quad}{3} \quad \frac{\quad}{1}$	$\frac{\quad}{2} \quad \frac{\quad}{2}$	$\frac{\quad}{3} \quad \frac{\quad}{3}$	$\frac{\quad}{4} \quad \frac{\quad}{3}$
l	$\frac{\quad}{3} \quad \frac{\quad}{3}$	$\frac{\quad}{4} \quad \frac{\quad}{2}$	$\frac{\quad}{3} \quad \frac{\quad}{2}$		
o	$\frac{\quad}{4} \quad \frac{\quad}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{1}{2} \quad \frac{2}{1}$	$\frac{2}{2} \quad \frac{3}{2}$	$\frac{2}{3} \quad \frac{4}{2}$	$\frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{1}{3} \quad \frac{2}{1}$	$\frac{2}{2} \quad \frac{3}{2}$	$\frac{3}{3} \quad \frac{3}{3}$	$\frac{3}{4} \quad \frac{4}{3}$
l	$\frac{\quad}{\quad} \frac{3}{3}$	$\frac{3}{4} \quad \frac{2}{2}$	$\frac{2}{3} \quad \frac{3}{2}$	$\frac{3}{3} \quad \frac{4}{?}$	
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{4}{3}$
l	$\frac{\quad}{\quad} \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{2}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{4}{3}$	
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{4}{3}$
l	$\frac{\quad}{\quad} \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{2}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{4}{3}$	$\frac{\quad}{\quad} \frac{4}{4} \quad \frac{4}{?}$
o	$\frac{\quad}{\quad} \frac{4}{4}$				

		s	n	o	w
	$\frac{\quad}{0}$	$\frac{1}{1}$	$\frac{2}{2}$	$\frac{3}{3}$	$\frac{4}{4}$
o	$\frac{\quad}{1}$	$\frac{1}{2}$	$\frac{2}{2}$	$\frac{2}{3}$	$\frac{4}{3}$
s	$\frac{\quad}{2}$	$\frac{1}{3}$	$\frac{2}{2}$	$\frac{3}{3}$	$\frac{3}{4}$
l	$\frac{\quad}{3}$	$\frac{3}{4}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{4}$
o	$\frac{\quad}{4}$				

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 1 \quad 2 \quad 2 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 2 \quad 4 \quad 3 \quad 2$	$\frac{\quad}{\quad} 4 \quad 5 \quad 3 \quad 3$
s	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 1 \quad 2 \quad 3 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3 \quad 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 4 \quad 4 \quad 3$
l	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 2 \quad 4 \quad 2$	$\frac{\quad}{\quad} 2 \quad 3 \quad 3 \quad 2$	$\frac{\quad}{\quad} 3 \quad 4 \quad 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4 \quad 4 \quad 4$
o	$\frac{\quad}{\quad} 4 \quad 4$	$\frac{\quad}{\quad} 4 \quad 3 \quad 5 \quad ?$			

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 1 \quad 2 \quad 2 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 2 \quad 4 \quad 3 \quad 2$	$\frac{\quad}{\quad} 4 \quad 5 \quad 3 \quad 3$
s	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 1 \quad 2 \quad 3 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3 \quad 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 4 \quad 4 \quad 3$
l	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 2 \quad 4 \quad 2$	$\frac{\quad}{\quad} 2 \quad 3 \quad 3 \quad 2$	$\frac{\quad}{\quad} 3 \quad 4 \quad 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4 \quad 4 \quad 4$
o	$\frac{\quad}{\quad} 4 \quad 4$	$\frac{\quad}{\quad} 4 \quad 3 \quad 5 \quad 3$			

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 1 \quad 2 \quad 2 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 2 \quad 4 \quad 3 \quad 2$	$\frac{\quad}{\quad} 4 \quad 5 \quad 3 \quad 3$
s	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 1 \quad 2 \quad 3 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3 \quad 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 4 \quad 4 \quad 3$
l	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 2 \quad 4 \quad 2$	$\frac{\quad}{\quad} 2 \quad 3 \quad 3 \quad 2$	$\frac{\quad}{\quad} 3 \quad 4 \quad 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4 \quad 4 \quad 4$
o	$\frac{\quad}{\quad} 4 \quad 4$	$\frac{\quad}{\quad} 4 \quad 3 \quad 5 \quad 3$	$\frac{\quad}{\quad} 3 \quad 3 \quad 4 \quad ?$		

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 1 \quad 2 \quad 2 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 2 \quad 4 \quad 3 \quad 2$	$\frac{\quad}{\quad} 4 \quad 5 \quad 3 \quad 3$
s	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 1 \quad 2 \quad 3 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3 \quad 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 4 \quad 4 \quad 3$
l	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 2 \quad 4 \quad 2$	$\frac{\quad}{\quad} 2 \quad 3 \quad 3 \quad 2$	$\frac{\quad}{\quad} 3 \quad 4 \quad 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4 \quad 4 \quad 4$
o	$\frac{\quad}{\quad} 4 \quad 4$	$\frac{\quad}{\quad} 4 \quad 3 \quad 5 \quad 3$	$\frac{\quad}{\quad} 3 \quad 3 \quad 4 \quad 3$		

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 1 \quad 2 \quad 2 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 2 \quad 4 \quad 3 \quad 2$	$\frac{\quad}{\quad} 4 \quad 5 \quad 3 \quad 3$
s	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 1 \quad 2 \quad 3 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3 \quad 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 4 \quad 4 \quad 3$
l	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 2 \quad 4 \quad 2$	$\frac{\quad}{\quad} 2 \quad 3 \quad 3 \quad 2$	$\frac{\quad}{\quad} 3 \quad 4 \quad 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4 \quad 4 \quad 4$
o	$\frac{\quad}{\quad} 4 \quad 4$	$\frac{\quad}{\quad} 4 \quad 3 \quad 5 \quad 3$	$\frac{\quad}{\quad} 3 \quad 3 \quad 4 \quad 3$	$\frac{\quad}{\quad} 2 \quad 4 \quad 4 \quad ?$	

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 1 \quad 2 \quad 2 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 2 \quad 4 \quad 3 \quad 2$	$\frac{\quad}{\quad} 4 \quad 5 \quad 3 \quad 3$
s	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 1 \quad 2 \quad 3 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3 \quad 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 4 \quad 4 \quad 3$
l	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 2 \quad 4 \quad 2$	$\frac{\quad}{\quad} 2 \quad 3 \quad 3 \quad 2$	$\frac{\quad}{\quad} 3 \quad 4 \quad 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4 \quad 4 \quad 4$
o	$\frac{\quad}{\quad} 4 \quad 4$	$\frac{\quad}{\quad} 4 \quad 3 \quad 5 \quad 3$	$\frac{\quad}{\quad} 3 \quad 3 \quad 4 \quad 3$	$\frac{\quad}{\quad} 2 \quad 4 \quad 4 \quad 2$	

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} \frac{1}{1}$	$\frac{\quad}{\quad} \frac{1}{2} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{3}$
s	$\frac{\quad}{\quad} \frac{2}{2}$	$\frac{\quad}{\quad} \frac{1}{3} \quad \frac{2}{1}$	$\frac{\quad}{\quad} \frac{2}{2} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{4}{3}$
l	$\frac{\quad}{\quad} \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{2}{2}$	$\frac{\quad}{\quad} \frac{2}{3} \quad \frac{3}{2}$	$\frac{\quad}{\quad} \frac{3}{3} \quad \frac{4}{3}$	$\frac{\quad}{\quad} \frac{4}{4} \quad \frac{4}{4}$
o	$\frac{\quad}{\quad} \frac{4}{4}$	$\frac{\quad}{\quad} \frac{4}{5} \quad \frac{3}{3}$	$\frac{\quad}{\quad} \frac{3}{4} \quad \frac{3}{3}$	$\frac{\quad}{\quad} \frac{2}{4} \quad \frac{4}{2}$	$\frac{\quad}{\quad} \frac{4}{3} \quad \frac{5}{?}$

		s	n	o	w
	$\frac{\quad}{\quad} 0$	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4$
o	$\frac{\quad}{\quad} 1 \quad 1$	$\frac{\quad}{\quad} 1 \quad 2 \quad 2 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 2 \quad 4 \quad 3 \quad 2$	$\frac{\quad}{\quad} 4 \quad 5 \quad 3 \quad 3$
s	$\frac{\quad}{\quad} 2 \quad 2$	$\frac{\quad}{\quad} 1 \quad 2 \quad 3 \quad 1$	$\frac{\quad}{\quad} 2 \quad 3 \quad 2 \quad 2$	$\frac{\quad}{\quad} 3 \quad 3 \quad 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 4 \quad 4 \quad 3$
l	$\frac{\quad}{\quad} 3 \quad 3$	$\frac{\quad}{\quad} 3 \quad 2 \quad 4 \quad 2$	$\frac{\quad}{\quad} 2 \quad 3 \quad 3 \quad 2$	$\frac{\quad}{\quad} 3 \quad 4 \quad 3 \quad 3$	$\frac{\quad}{\quad} 4 \quad 4 \quad 4 \quad 4$
o	$\frac{\quad}{\quad} 4 \quad 4$	$\frac{\quad}{\quad} 4 \quad 3 \quad 5 \quad 3$	$\frac{\quad}{\quad} 3 \quad 3 \quad 4 \quad 3$	$\frac{\quad}{\quad} 2 \quad 4 \quad 4 \quad 2$	$\frac{\quad}{\quad} 4 \quad 5 \quad 3 \quad 3$

Tabulate the trace

		s	n	o	w
	<u> </u> 0	<u> 1 </u> 1	<u> 2 </u> 2	<u> 3 </u> 3	<u> 4 </u> 4
o	<u> 1 </u> 1	<u> 1 2 </u> 2 1	<u> 2 3 </u> 2 2	<u> 2 4 </u> 3 2	<u> 4 5 </u> 3 3
s	<u> 2 </u> 2	<u> 1 2 </u> 3 1	<u> 2 3 </u> 2 2	<u> 3 3 </u> 3 3	<u> 3 4 </u> 4 3
l	<u> 3 </u> 3	<u> 3 2 </u> 4 2	<u> 2 3 </u> 3 2	<u> 3 4 </u> 3 3	<u> 4 4 </u> 4 4
o	<u> 4 </u> 4	<u> 4 3 </u> 5 3	<u> 3 3 </u> 4 3	<u> 2 4 </u> 4 2	<u> 4 5 </u> 3 3

		s	n	o	w
	0	1 1	2 2	3 3	4 4
o	1 1	1 2 2 1	2 3 2 2	2 4 3 2	4 5 3 3
s	2 2	1 2 3 1	2 3 2 2	3 3 3 3	3 4 4 3
l	3 3	3 2 4 2	2 3 3 2	3 4 3 3	4 4 4 4
o	4 4	4 3 5 3	3 3 4 3	2 4 4 2	4 5 3 3

How do I read out the editing operations that transform OSLO into SNOW?

		s	n	o	w
	0	1 1	2 2	3 3	4 4
o	1 1	1 2 2 1	2 3 2 2	2 4 3 2	4 5 3 3
s	2 2	1 2 3 1	2 3 2 2	3 3 3 3	3 4 4 3
l	3 3	3 2 4 2	2 3 3 2	3 4 3 3	4 4 4 4
o	4 4	4 3 5 3	3 3 4 3	2 4 4 2	4 5 3 3

cost	operation	input	output
1	insert	*	w

		s	n	o	w
	0	1 1	2 2	3 3	4 4
o	1 1	1 2 2 1	2 3 2 2	2 4 3 2	4 5 3 3
s	2 2	1 2 3 1	2 3 2 2	3 3 3 3	3 4 4 3
l	3 3	3 2 4 2	2 3 3 2	3 4 3 3	4 4 4 4
o	4 4	4 3 5 3	3 3 4 3	2 4 4 2	4 5 3 3

cost	operation	input	output
0	(copy)	o	o
1	insert	*	w

		s	n	o	w
	0	1 1	2 2	3 3	4 4
o	1 1	1 2 2 1	2 3 2 2	2 4 3 2	4 5 3 3
s	2 2	1 2 3 1	2 3 2 2	3 3 3 3	3 4 4 3
l	3 3	3 2 4 2	2 3 3 2	3 4 3 3	4 4 4 4
o	4 4	4 3 5 3	3 3 4 3	2 4 4 2	4 5 3 3

cost	operation	input	output
1	replace	l	n
0	(copy)	o	o
1	insert	*	w

		s	n	o	w
	0	1 1	2 2	3 3	4 4
o	1 1	1 2 2 1	2 3 2 2	2 4 3 2	4 5 3 3
s	2 2	1 2 3 1	2 3 2 2	3 3 3 3	3 4 4 3
l	3 3	3 2 4 2	2 3 3 2	3 4 3 3	4 4 4 4
o	4 4	4 3 5 3	3 3 4 3	2 4 4 2	4 5 3 3

cost	operation	input	output
0	(copy)	s	s
1	replace	l	n
0	(copy)	o	o
1	insert	*	w

		s	n	o	w
	0	1 1	2 2	3 3	4 4
o	1 1	1 2 2 1	2 3 2 2	2 4 3 2	4 5 3 3
s	2 2	1 2 3 1	2 3 2 2	3 3 3 3	3 4 4 3
l	3 3	3 2 4 2	2 3 3 2	3 4 3 3	4 4 4 4
o	4 4	4 3 5 3	3 3 4 3	2 4 4 2	4 5 3 3

cost	operation	input	output
1	delete	o	*
0	(copy)	s	s
1	replace	l	n
0	(copy)	o	o
1	insert	*	w

oslo → *slo* → *slo* → *sno* → *sno* → *snow*

There can be multiple optimal solutions

		c		a		t		c		a		t	
		0	1 1	2 2	3 3	4 4	5 5	6 6					
c		1	0 2	2 3	3 4	3 5	5 6	6 7					
		1	2 0	1 1	2 2	3 3	4 4	5 5					
a		2	2 1	0 2	2 3	3 4	3 5	5 6					
		2	3 1	2 0	1 1	2 2	3 3	4 4					
t		3	3 2	2 1	0 2	2 3	3 4	3 5					
		3	4 2	3 1	2 0	1 1	2 2	3 3					

		c	a	t	c	a	t
	0	1 1	2 2	3 3	4 4	5 5	6 6
c	1 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4	6 7 5 5
a	2 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4
t	3 3	3 2 4 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3

cost	operation	input	output
1	insert	*	c
1	insert	*	a
1	insert	*	t
0	(copy)	c	c
0	(copy)	a	a
0	(copy)	t	t

		c	a	t	c	a	t
	0	1 1	2 2	3 3	4 4	5 5	6 6
c	1 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4	6 7 5 5
a	2 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4
t	3 3	3 2 4 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3

cost	operation	input	output
0	(copy)	c	c
1	insert	*	a
1	insert	*	t
1	insert	*	c
0	(copy)	a	a
0	(copy)	t	t

		c	a	t	c	a	t
	0	1 1	2 2	3 3	4 4	5 5	6 6
c	1 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4	6 7 5 5
a	2 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4
t	3 3	3 2 4 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3

cost	operation	input	output
0	(copy)	c	c
0	(copy)	a	a
1	insert	*	t
1	insert	*	c
1	insert	*	a
0	(copy)	t	t

		c		a		t		c		a		t	
		0	1 1	2 2	3 3	4 4	5 5	6 6					
c		1	0 2	2 3	3 4	3 5	5 6	6 7					
		1	2 0	1 1	2 2	3 3	4 4	5 5					
a		2	2 1	0 2	2 3	3 4	3 5	5 6					
		2	3 1	2 0	1 1	2 2	3 3	4 4					
t		3	3 2	2 1	0 2	2 3	3 4	3 5					
		3	4 2	3 1	2 0	1 1	2 2	3 3					

cost	operation	input	output
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		c	a	t	c	a	t
	0	1 1	2 2	3 3	4 4	5 5	6 6
c	1 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4	6 7 5 5
a	2 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4
t	3 3	3 2 4 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3

cost	operation	input	output
0	(copy)	c	c

		c		a		t		c		a		t		
		0	1	1	2	2	3	3	4	4	5	5	6	6
c		1	0	2	2	3	3	4	3	5	5	6	6	7
		1	2	0	1	1	2	2	3	3	4	4	5	5
a		2	2	1	0	2	3	4	3	4	3	5	5	6
		2	3	1	2	0	1	1	2	2	3	3	4	4
t		3	3	2	2	0	2	3	2	3	3	4	3	5
		3	4	2	3	1	2	0	1	1	2	2	3	3

cost	operation	input	output
0	(copy)	c	c
0	(copy)	a	a

		c	a	t	c	a	t
	0	1 1	2 2	3 3	4 4	5 5	6 6
c	1 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4	6 7 5 5
a	2 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4
t	3 3	3 2 4 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3

cost	operation	input	output
0	(copy)	c	c
0	(copy)	a	a
0	(copy)	t	t

		c		a		t		c		a		t		
		0	1	1	2	2	3	3	4	4	5	5	6	6
c		1	0	2	2	3	3	4	3	5	5	6	6	7
		1	2	0	1	1	2	2	3	3	4	4	5	5
a		2	2	1	0	2	2	3	3	4	3	5	5	6
		2	3	1	2	0	1	1	2	2	3	3	4	4
t		3	3	2	2	1	0	2	2	3	3	4	3	5
		3	4	2	3	1	2	0	1	1	2	2	3	3

cost	operation	input	output
0	(copy)	c	c
0	(copy)	a	a
0	(copy)	t	t

		c	a	t	c	a	t
	0	1 1	2 2	3 3	4 4	5 5	6 6
c	1 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4	6 7 5 5
a	2 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4
t	3 3	3 2 4 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3

cost	operation	input	output
0	(copy)	c	c
0	(copy)	a	a
0	(copy)	t	t
1	insert	*	c

		c	a	t	c	a	t
	0	1 1	2 2	3 3	4 4	5 5	6 6
c	1 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4	6 7 5 5
a	2 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4
t	3 3	3 2 4 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3

cost	operation	input	output
0	(copy)	c	c
0	(copy)	a	a
0	(copy)	t	t
1	insert	*	c
1	insert	*	a

		c	a	t	c	a	t
	0	1 1	2 2	3 3	4 4	5 5	6 6
c	1 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4	6 7 5 5
a	2 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3	5 6 4 4
t	3 3	3 2 4 2	2 1 3 1	0 2 2 0	2 3 1 1	3 4 2 2	3 5 3 3

cost	operation	input	output
0	(copy)	c	c
0	(copy)	a	a
0	(copy)	t	t
1	insert	*	c
1	insert	*	a
1	insert	*	t

Takeways

- Dynamic programming solves problems by combining solutions to subproblems
- In this sense it is similar to Divide-and-Conquer, but the sub-problems are not independent
- Divide-and-Conquer does more work than necessary, repeatedly solving common subproblems
- Dynamic programming solves every subproblem just once and saves its answer in a table.
- 'Programming' here refers to a tabular method, not writing code
- It is typically applied to optimization problems.